

$$\frac{\Gamma \vdash p_1 : t_1 \rightarrow t_2 \quad \Gamma \vdash p_2 : t_1}{\Gamma \vdash p_1 p_2 : t_2} \text{ (T-APP)}$$

$$\frac{\text{① } \begin{array}{l} \Gamma, p_1, p_2, t_1, t_2 \notin \text{FV}(\Gamma^\circ, e^\circ, T^\circ) \\ \text{Unify} \left(\begin{array}{l} \Gamma = \Gamma^\circ \\ p_1 p_2 = e^\circ \\ t_2 = T^\circ \end{array} \right) = \sigma \end{array} \quad \begin{array}{l} \text{② } \sigma(\Gamma) \vdash \sigma(p_1) : \sigma(t_1 \rightarrow t_2) \rightsquigarrow \sigma_1 \\ \text{③ } \sigma' = \sigma_1 \sigma \\ \sigma'(\Gamma) \vdash \sigma'(p_2) : \sigma'(t_1) \rightsquigarrow \sigma_2 \end{array}}{\Gamma^\circ \vdash e^\circ : T^\circ \rightsquigarrow \sigma_2 \sigma' |_{\text{FV}(\Gamma^\circ, e^\circ, T^\circ)}} \text{ (S-APP)}$$